

Title

Pyrogenic Organic Matter Laboratory Experiment: Aerobic Respiration and Geochemistry from Variably Inundated Stream Sediments (v2)

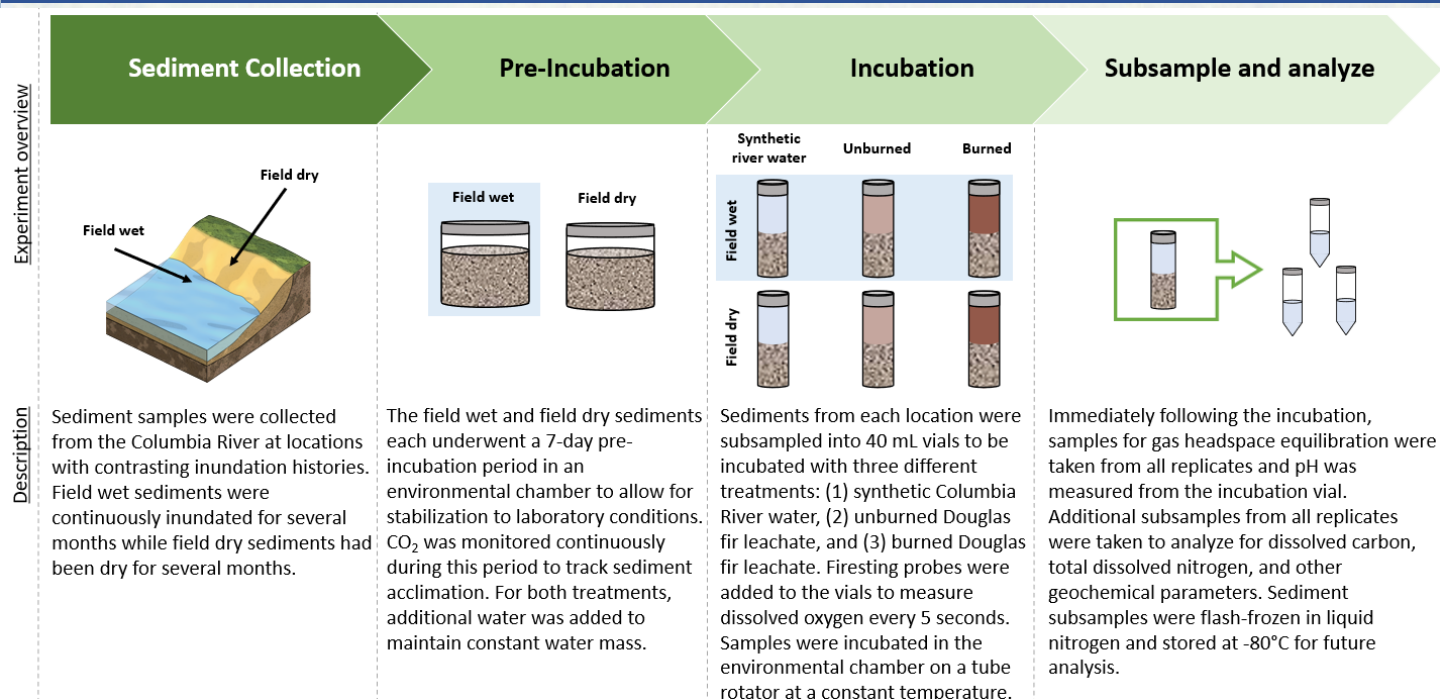
Summary

This dataset supports a broader study examining the effects of variable inundation and pyrogenic organic matter on ecosystem respiration. The dataset provides data generated from a laboratory batch experiment investigating the interaction between variable inundation conditions (wet and dry sediment) and pyrogenic organic matter (burned and unburned treatments). The contents include time series dissolved oxygen, sediment geochemistry data, and field metadata (including qualitative information on instream and river corridor characteristics).

This data package was originally published in November 2025. It was updated in April 2026 (v2; new and modified files). See the change history section below for more details.

For details on how to navigate data packages generated by this project, see <https://data.ess-dive.lbl.gov/portals/PNNLRiverCorridorSFA/About>.

Experimental Design



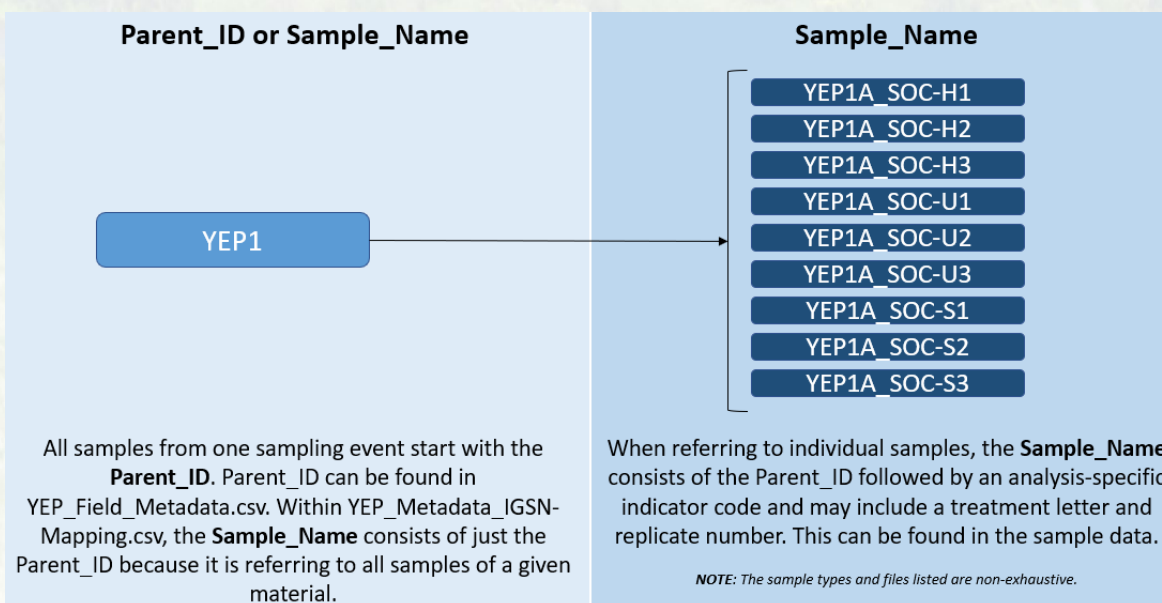
Brief Overview of Methods

Sediment samples were collected from the Columbia River at locations with contrasting inundation conditions – field wet sediments that had been continuously inundated for several months and field dry sediments from a higher elevation on the same shoreline that had been dry for several months. Site metadata, including general environmental information, were manually recorded in the field. See YEP_Field_Protocol.pdf for details on sample collection and site metadata collection. Briefly, sediment samples were sieved to less than 2 millimeters and transported on blue ice to Pacific Northwest National Laboratory where they underwent a 7-day pre-incubation period in an environmental chamber to allow

stabilization to laboratory conditions. After the pre-incubation period, sediment incubations were conducted using three treatments: (1) synthetic Columbia River water (0 mg C/L) as a negative control, (2) unburned Douglas fir leachate at 9 mg C/L, and (3) burned (high severity) Douglas fir leachate at 9 mg C/L. All treatments were applied to both field wet and field dry sediments. Aerobic respiration rates were calculated from the slope of the linear regression between dissolved oxygen data measured using FireSting oxygen probes and incubation time. Incubations were monitored continuously with measurements every 5 seconds, with experiments ending after 6 hours or when dissolved oxygen levels reached 2.0 mg/L. For details regarding laboratory and instrument methods, see the alphanumeric methods codes located in the header rows of the chemistry data csv file and their associated definitions in *YEP_Methods_Codes.csv. Data formats follow ESS-DIVE Reporting Formats.

Critical Details

- 1 – **Sample Name:** The sample names contain pertinent information for understanding the data (i.e., material type, treatment, replicate), see “v*_YEP_Sample_Name_Metadata.csv” to understand the details of each sample name. *Note: the sample names in the file may go beyond what is published in this data package.*
- 2 – **Sample Name Mapping:** Each sampling event has a unique Parent_ID in the format YEP#. The field metadata and data files all contain these unique IDs and can be mapped across each other accordingly. The Parent_ID may have other indicators appended when referring to samples. See figure below for details.



- 3 – **File naming:** The files within this data package follow a consistent naming scheme to enable easy use of the data and support AI-readiness. All files begin with “YEP” to indicate the study. The text following this indicator, and separated with an underscore, describe the contents, material, and/or data type(s) within the file. The primary sample material in this data package is sediment; however, some files with “Sediment” in the filename contain data where the material is listed as “Liquid>aqueous.” within the file. These samples contain either the treatment solution without any sediment present or Milli-Q water for process blanks (see “v*_YEP_Sample_Name_Metadata.csv”

for more details on the samples). Additionally, a version number may be at the beginning of the file name. See the change history section below.

- 4 – **File Joining:** The file-level metadata (FLMD) file includes a column that lists the fields in each file to facilitate joining. Each sample name ends with a replicate number, where each replicate indicates an individual reactor. To merge the data files across different data types, match samples based on their replicate number and material, excluding the analyte code. Example: Data from YEP1A_GAS-S1 can be merged with YEP1A_INC-S1 because they share the same base name (YEP1A), replicate number (S1), and material (sediment) — only the analyte codes (GAS vs. INC) differ. Note that material type is not explicitly listed in the sample name, but it can be pulled from the sample data file “Material” columns, which uses a controlled vocabulary.
- 5 – **Replicate Numbers:** Replicates were incubated on different days after field sample collection. In most cases, replicates 1 and 2 were incubated on Day 1, replicate 3 was incubated on Day 2, and replicates 4 and 5 were incubated on Day 3. However, the burned replicate 1 (-H1) and the unburned replicate 2 (-U2) were incubated on Day 2.
- 6 – **Source Leachates:** The leachates used for the treatments were created using leftover material from the BSLE_0013 and BSLE_0050 Douglas fir chars (Grieger et al., 2022). Metadata about the chars and methods used to create the leachates can be found at <https://data.ess-dive.lbl.gov/view/doi%3A10.15485%2F1894135> (Grieger et al., 2022).
- 7 – **Partial Pressure Calculation:** Partial pressure of CO₂, CH₄, and N₂O was converted to molar units using the temperature-dependent Henry's Law constant (K_H) as seen in Regier et al. (2023):

$$K_H = e(A1+A2*(100/T)+A3*\ln(T/100)+S*(B1+B2*(T/100)+B3*(T/100)^2))$$

where A1-3 and B1-3 are gas specific coefficients for the temperature (T) and salinity (S) dependence of solubilities. All incubations were conducted at 21°C, with an estimated salinity of 0.1 PSU. K_H was multiplied by the partial pressure of the gas (p) to get the dissolved molar concentration (C):

$$C = K_H * p$$

- 8 – **FTICR-MS processing levels:** FTICR-MS data is provided in three processing levels in this data package. The unprocessed instrument files are available from the Environmental Molecular Sciences Laboratory (EMSL). These can be processed in newly developed public workflows (i.e., CoreMS) for data use or they can be first processed in a proprietary software (i.e., Bruker) into XML files and then processed in publicly available Formultitude (previously called Formularity; <https://github.com/PNNL-Comp-Mass-Spec/Formultitude>) for further data use. In previous data packages, we have processed the files in Bruker prior to publishing and then only published XML files and instructions for the user to process the XML files in Formultitude. In this data package, due to these new software developments, we have included pre-processed (i.e., peak picked) XML files, CoreMS output files, and fully processed data to allow for maximum data user flexibility
- 9 – **FTICR-MS processing scripts:** We have provided a static version of our CoreMS processing instructions and scripts that were used for this dataset. These instructions and scripts may change over time as this new pipeline evolves. The most up to date instructions and scripts can be found at

https://github.com/danczakre/RCSFA_CoreMS/tree/main/. A release was created on GitHub to document the version pulled for this data package.

- 10 – **FTICR-MS merging across datasets:** We do not recommend combining processed data across datasets processed separately, whether the data were run through Formultitude or CoreMS. If you wish to combine data from across datasets, you need to go back to at least the .xml files and reprocess the data using your pipeline of choice. For convenience, we have included the CoreMS outputs which **can** be combined with other CoreMS outputs released in our data packages, though this is not ideal. We are happy to work through these considerations with you if you have questions.
- 11 – **FTICR Blank Correction:** Given that we detected higher than expected assigned formulas in the blanks, we recommend blank subtraction (e.g., remove peaks identified in >50% of our blanks from the finalized dataset). We did not do blank subtraction in the data provided.

Data Package Structure

In addition to this readme, this data package also includes a file-level metadata (FLMD) file that describes each file and a data dictionary (DD) that describes all column/row headers and variable definitions.

This dataset is comprised of (1) file-level metadata; (2) data dictionary; (3) field metadata; (4) international generic sample number (IGSN) mapping file; (5) readme; (6) field protocol; (7) sample name metadata; (8) an environmental context picture for the dry and inundated sampling locations; and (9) a subfolder with sample data from the sediment incubation experiment. The sample data subfolder contains (1) dissolved organic carbon (DOC, measured as non-purgeable organic carbon, NPOC); (2) total nitrogen (TN); (3) gravimetric moisture; (4) partial pressure and production rates of carbon dioxide, methane, and nitrous oxide; (5) field wet sediment mass, dry sediment mass, water mass, and field wet sediment volume in incubation and sediment NPOC/TN vials; (6) methods codes; (7) respiration rates, raw time series dissolved oxygen and temperature, and a subfolder containing associated plots and scripts; (8) ions; (9) FTICR-MS methods; and (10) a subfolder of 12 Tesla (12T) FTICR-MS data. This folder contains the CoreMS processed data and three subfolders, one containing the .xml files, one containing the CoreMS output files, and the other containing instructions and scripts for processing the files in CoreMS (<https://github.com/EMSL-Computing/CoreMS>). All files are .csv, .pdf, .R, .xml, .html, .Rmd, .py, .cal, .json, or .jpg.

Acknowledgements, Citations, and License

Acknowledgments

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Citations

- Grieger S., et al. (2022): Organic matter concentration and composition of experimentally burned open air and muffle furnace vegetation chars across differing burn severity and feedstock types from Pacific Northwest, USA (v4). River Corridor and Watershed Biogeochemistry SFA, ESS-DIVE repository. Dataset. [doi:10.15485/1894135](https://doi.org/10.15485/1894135)
- Regier, P., et al. (2023). Coastal inundation regime moderates the short-term effects of sediment and soil additions on seawater oxygen and greenhouse gas dynamics: A microcosm experiment. *Frontiers in Marine Science*, 10, 1308590. <https://doi.org/10.3389/fmars.2023.1308590>

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Change History

Approach to change history and versioning:

Updates to **data package** version: When any file within a data package is updated, the data package version number is updated. The data package version number is indicated in the title of the data package, the data package folder name, and in the change history table below. You can access previous versions of the data package by sending a request to ESS-DIVE.

Updates to **individual file** versions: As files are changed, the file version number is also updated. The file version number is indicated in the file name, file level metadata (flmd) file, and the change history table below. The version number on an individual file may not match the version number of the data package. For example, v3 of a data package may include v2 of an individual file.

The change history below describes each file revised during versioning. If you are interested in seeing the exact cells within a file that have changed, you can utilize the daff package in R (<https://github.com/edwindj/daff>) to compare a previously downloaded file to a newly downloaded file.

In the change history table below, the sub-headers and bullets indicate the type of change in each file:

- New files: Describes new files added that were not present in previous data package versions.
- Bulk changes to files: Describes a change to many files within the data package. The indicated superscript will be added to each file name that the change applies to.
- Modified files:
 - Corrected: Describes existing information modified or removed to prevent sharing of incorrect information
 - Added: Describes new information inserted into an existing file (e.g., appending new columns/rows)
 - Updated: Describes modifying existing information to maintain accuracy though version changes. (e.g., changing version number to new version number)

Change history:

Data Package Version	Changes
Version 1 <i>November 2025</i>	Original data package publication
Version 2 <i>April 2026</i>	<u>NEW FILES</u> • YEP_Sediment_Ions.csv • YEP_Sediment_Incubations_Respiration_Rates.csv • YEP_Calculate_Normalized_Respiration.R • YEP_Calculate_Respiration.R • YEP_Sediment_Incubations_DO_vs_Incubation_Time_Plots.pdf • YEP_Sediment_FTICR_Methods.csv • YEP_*.xml • YEP_*.corems.csv • YEP_*.corems.json • YEP_*.corems.cal • YEP_Sediment_CoreMS_Processed_ICR_Calibration.csv • YEP_Sediment_CoreMS_Processed_ICR_Data.csv • YEP_Sediment_CoreMS_Processed_ICR_Mol.csv • CoreMS_MergeProcess.Rmd • CoreMS_Runner.py • Format_CoreMS_Outputs.R • getLambda.R • Instructions_for_Processing_FTICR_Data_using_CoreMS.pdf <u>MODIFIED FILES</u> YEP_Sediment_Water_Mass_Volume (v2) • Removed rows for samples that have no corresponding data generated YEP_Sediment_CO2_CH4_N2O.csv (v2) • Added columns for partial pressure of CH4 and N2O • Added columns of rates for CO2, CH4, and N2O • Updated file name from "YEP_Sediment_CO2.csv" YEP_Sample_Name_Metadata.csv (v2) • Added new samples YEP_Methods_Codes.csv (v2) • Updated description of carbon and nitrogen content for clarity YEP_flmd.csv (v2) • Added rows for new files • Updated version number for modified files. YEP_dd.csv (v2) • Added rows for new column headers from new files • Updated to reflect columns in modified files readme_YEP.pdf (v2) • Updated versioning information. • Updated version number in data package title

Note: An asterisks (*) in a file name indicates multiple files with the same file and file naming structure. See the flmd for more details about specific files and the meaning of the asterisks.